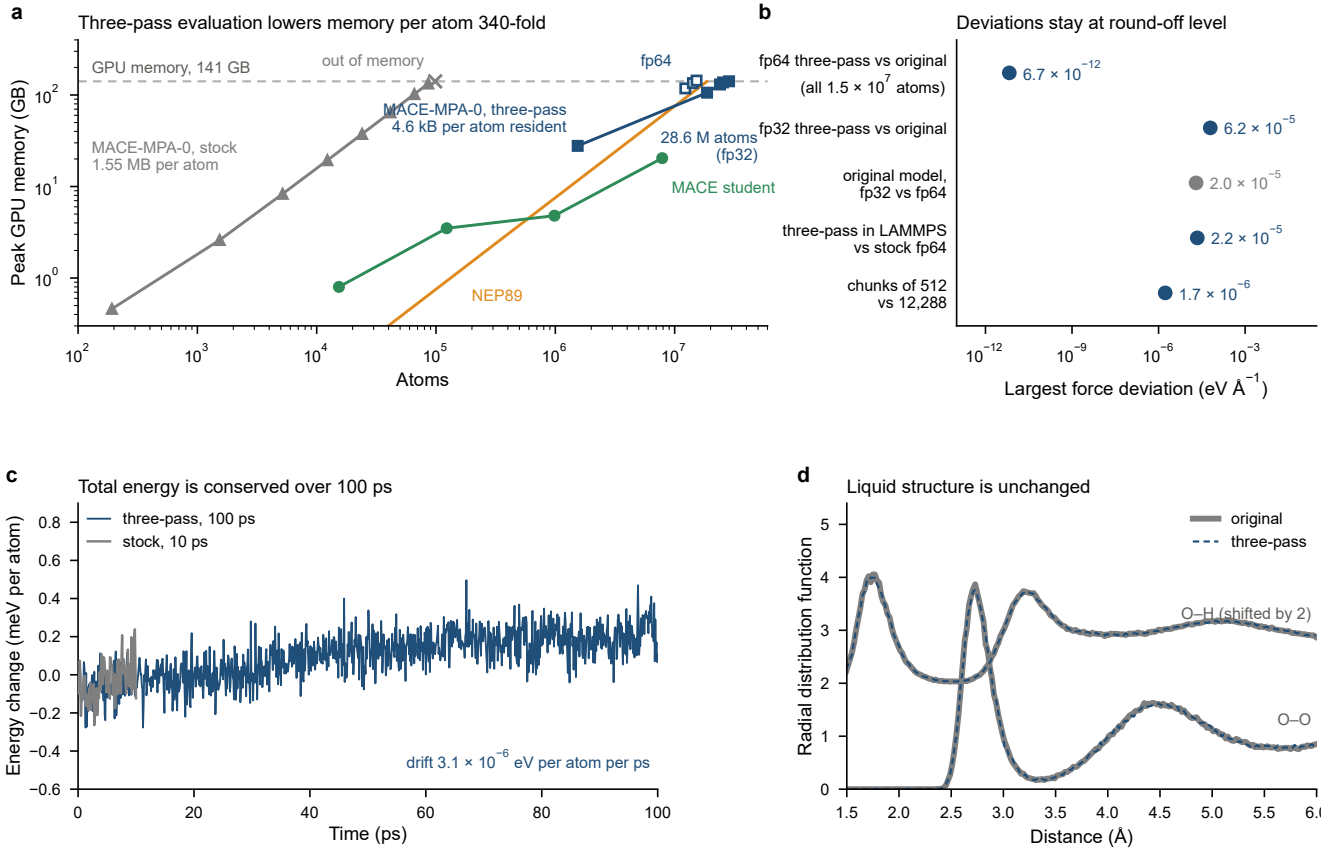

a
Three-pass evaluation lowers memory per atom 340-fold
GPU memory, 141 GB
out of memory
fp64
MACE-MPA-0, three-pass
4.6 kB per atom resident
MACE-MPA-0, stock
1.55 MB per atom
28.6 M atoms
(fp32)
MACE student
NEP89
Peak GPU memory (GB)
Atoms
b
Deviations stay at round-off level
fp64 three-pass vs original
(all 1.5 × 10^7 atoms)
6.7 × 10^−12
fp32 three-pass vs original
6.2 × 10^−5
original model,
fp32 vs fp64
2.0 × 10^−5
three-pass in LAMMPS
vs stock fp64
2.2 × 10^−5
chunks of 512
vs 12,288
1.7 × 10^−6
Largest force deviation (eV Å^−1)
c
Total energy is conserved over 100 ps
three-pass, 100 ps
stock, 10 ps
drift 3.1 × 10^−6 eV per atom per ps
Energy change (meV per atom)
Time (ps)
d
Liquid structure is unchanged
original
three-pass
O–H (shifted by 2)
O–O
Radial distribution function
Distance (Å)